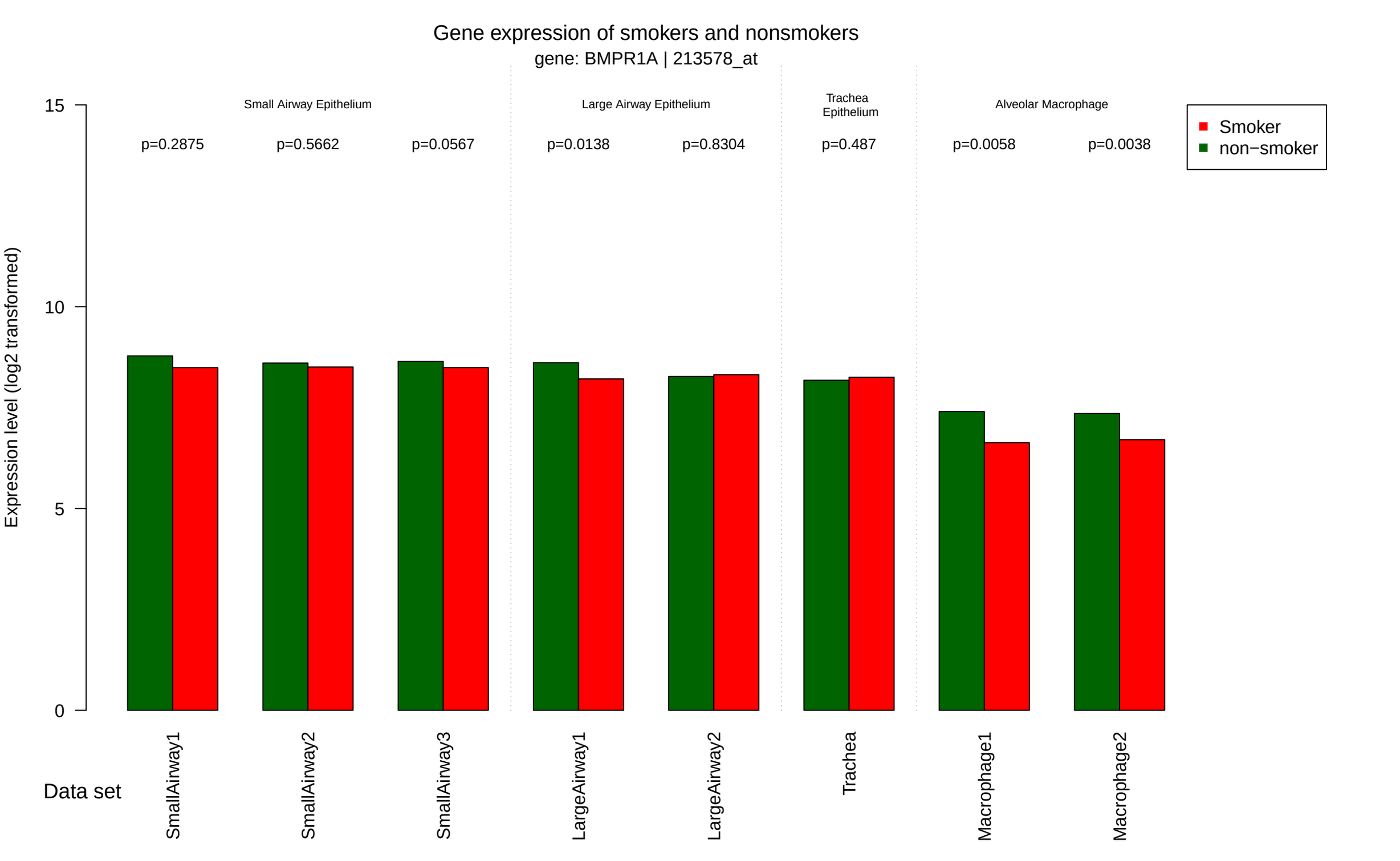
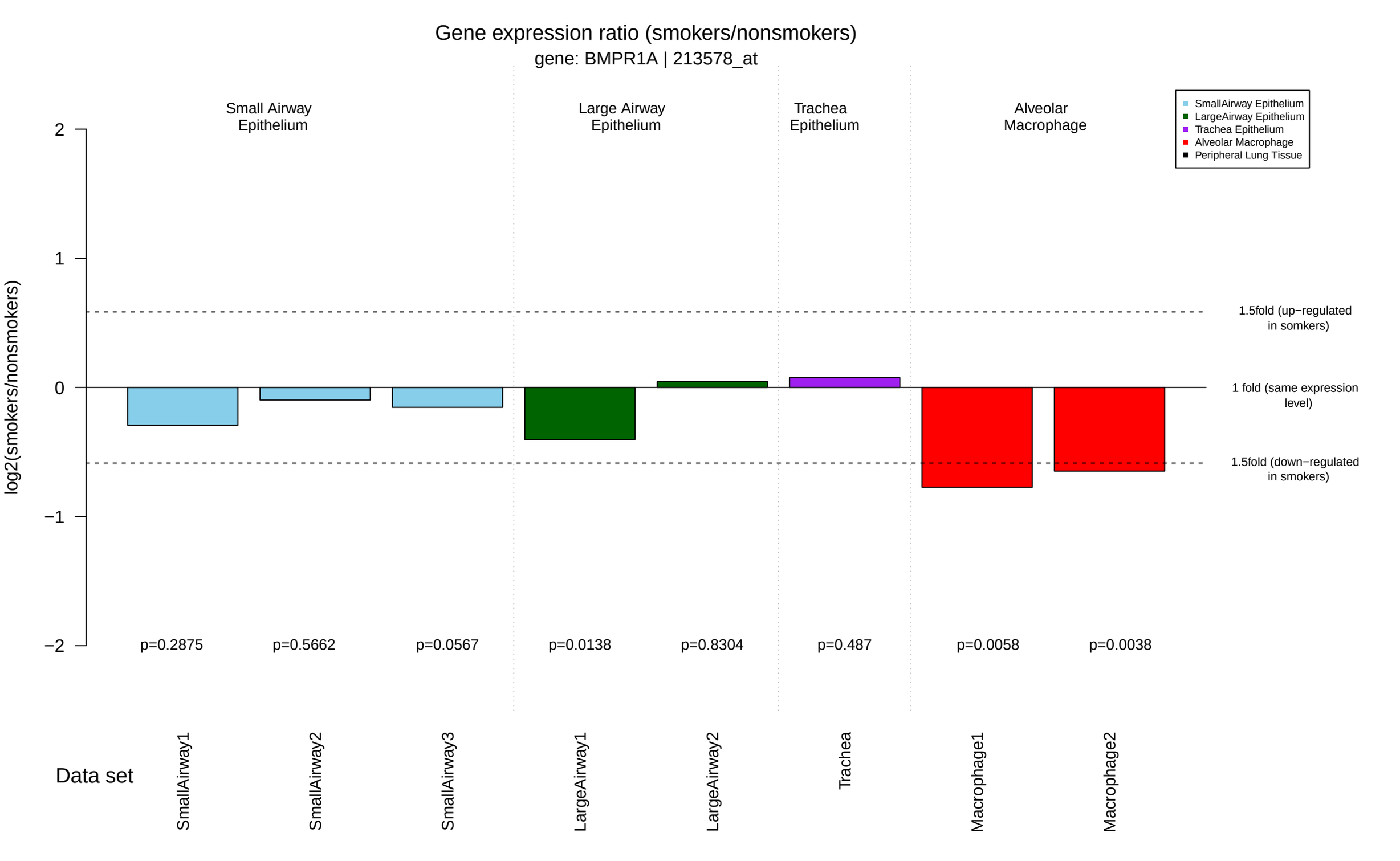
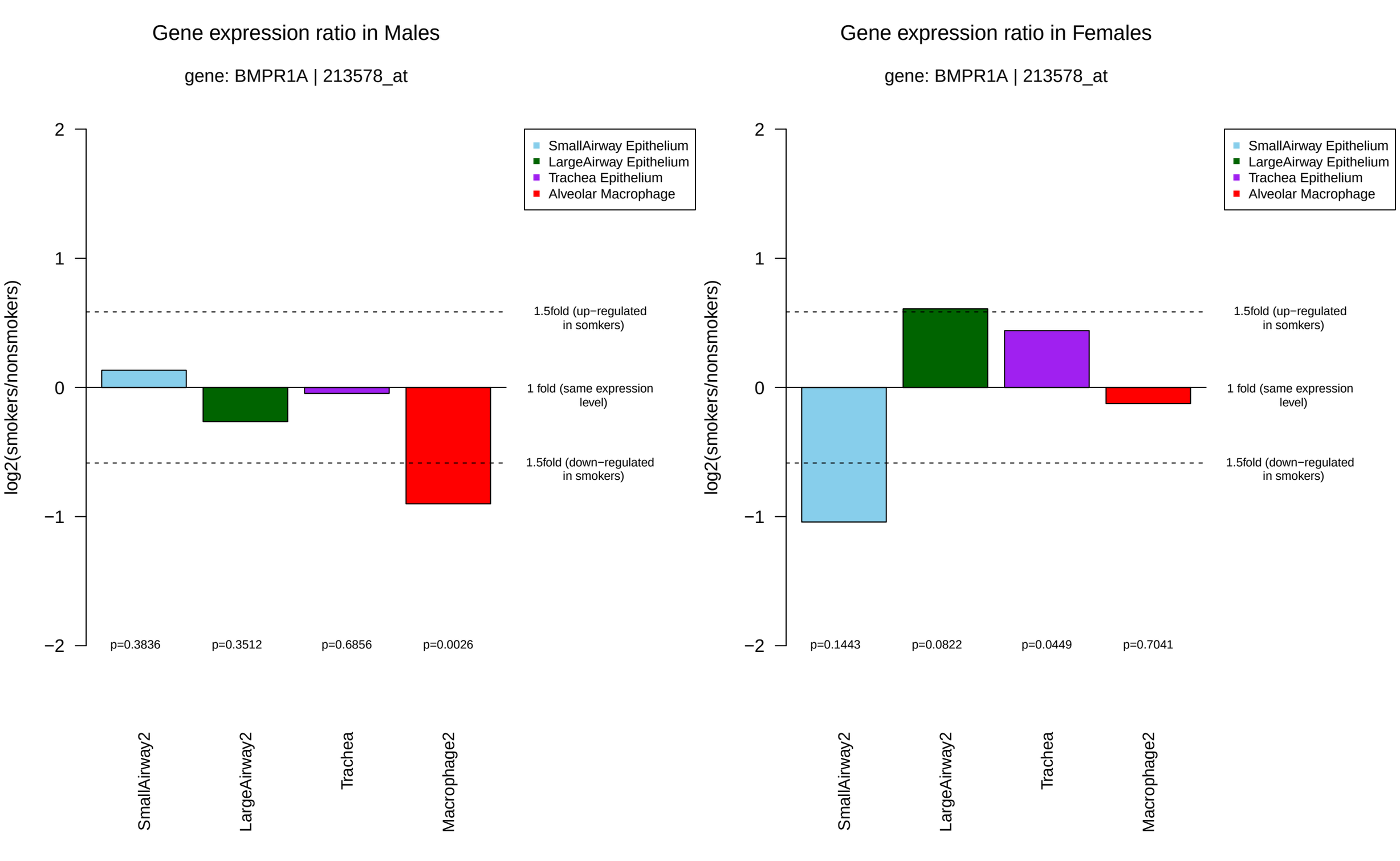
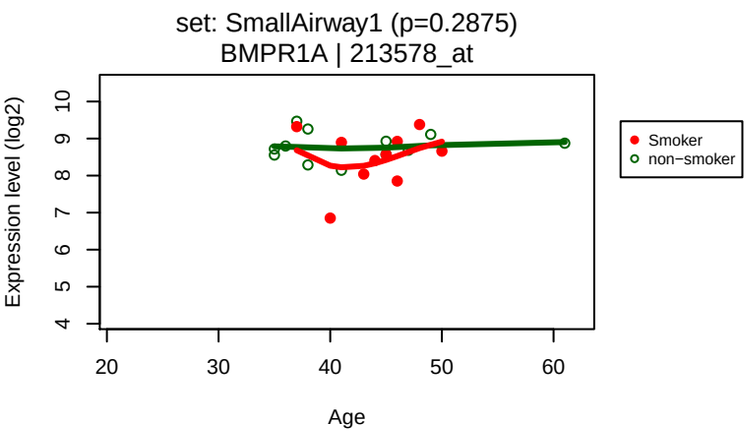
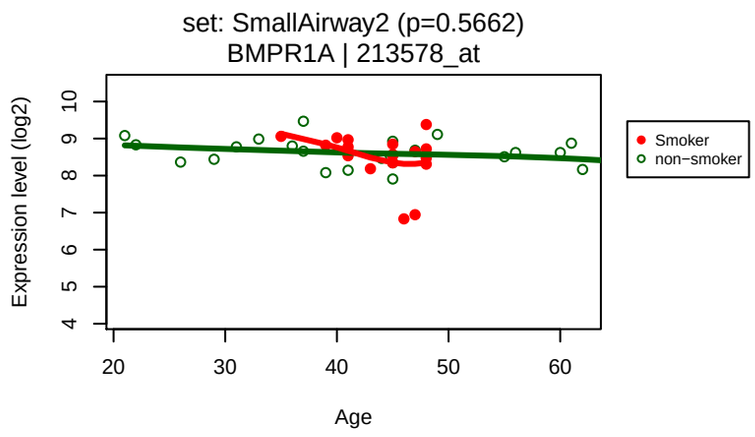
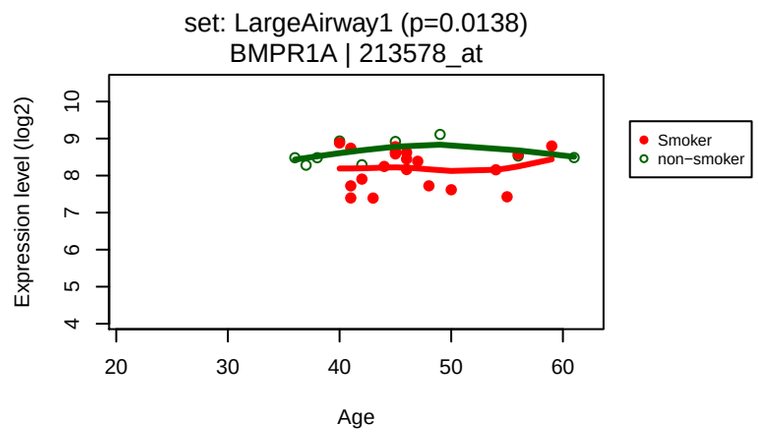

Gene expression of smokers and nonsmokers
gene: BMPR1A | 213578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2875
p=0.5662
p=0.0567
p=0.0138
p=0.8304
p=0.487
p=0.0058
p=0.0038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

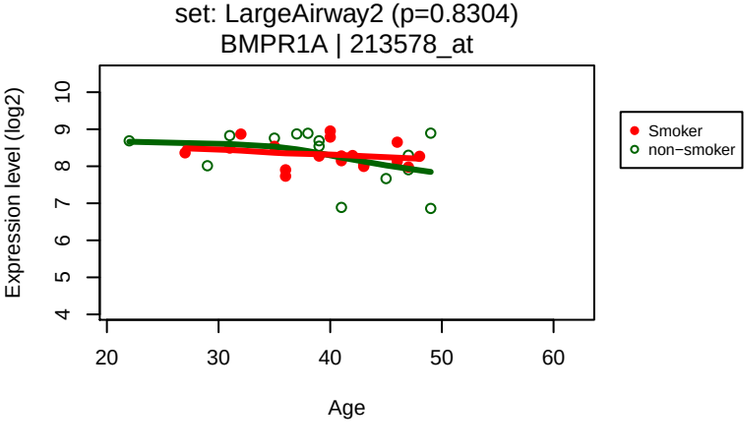

Gene expression ratio (smokers/nonsmokers)
gene: BMPR1A | 213578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2875
p=0.5662
p=0.0567
p=0.0138
p=0.8304
p=0.487
p=0.0058
p=0.0038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

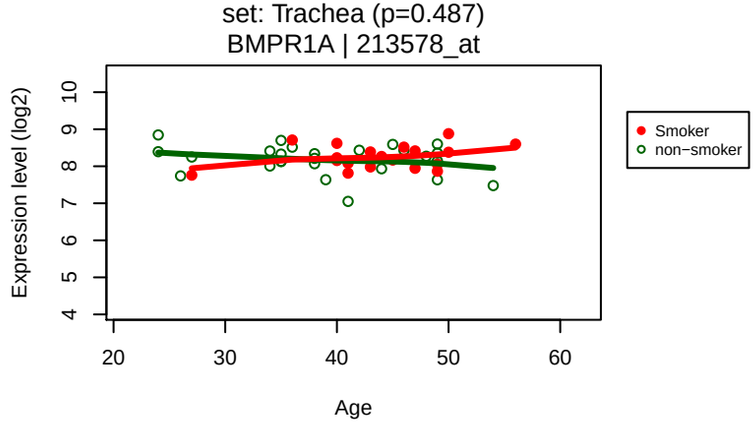

Gene expression ratio in Males
gene: BMPR1A | 213578_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3836
p=0.3512
p=0.6856
p=0.0026
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMPR1A | 213578_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1443
p=0.0822
p=0.0449
p=0.7041
SmallAirway2
LargeAirway2
Trachea
Macrophage2

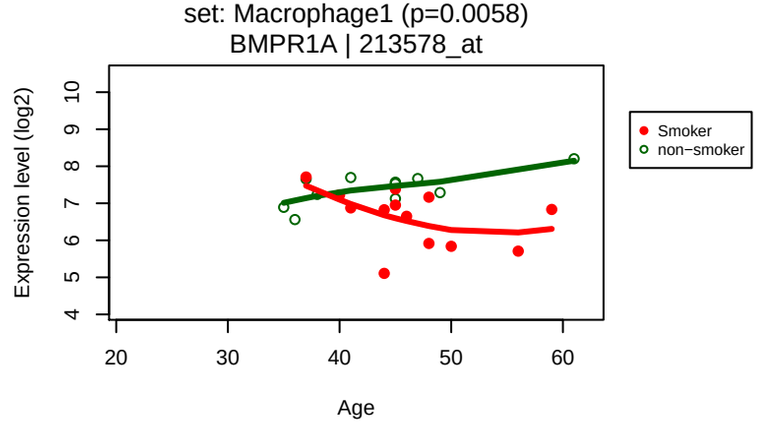

set: SmallAirway1 (p=0.2875)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker

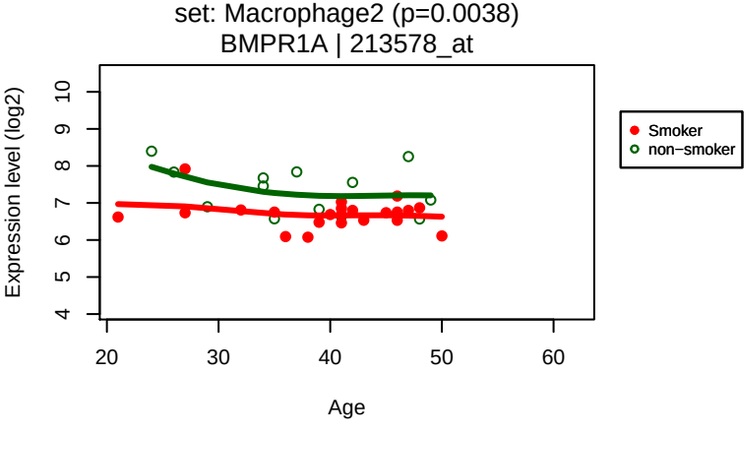

set: SmallAirway2 (p=0.5662)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker

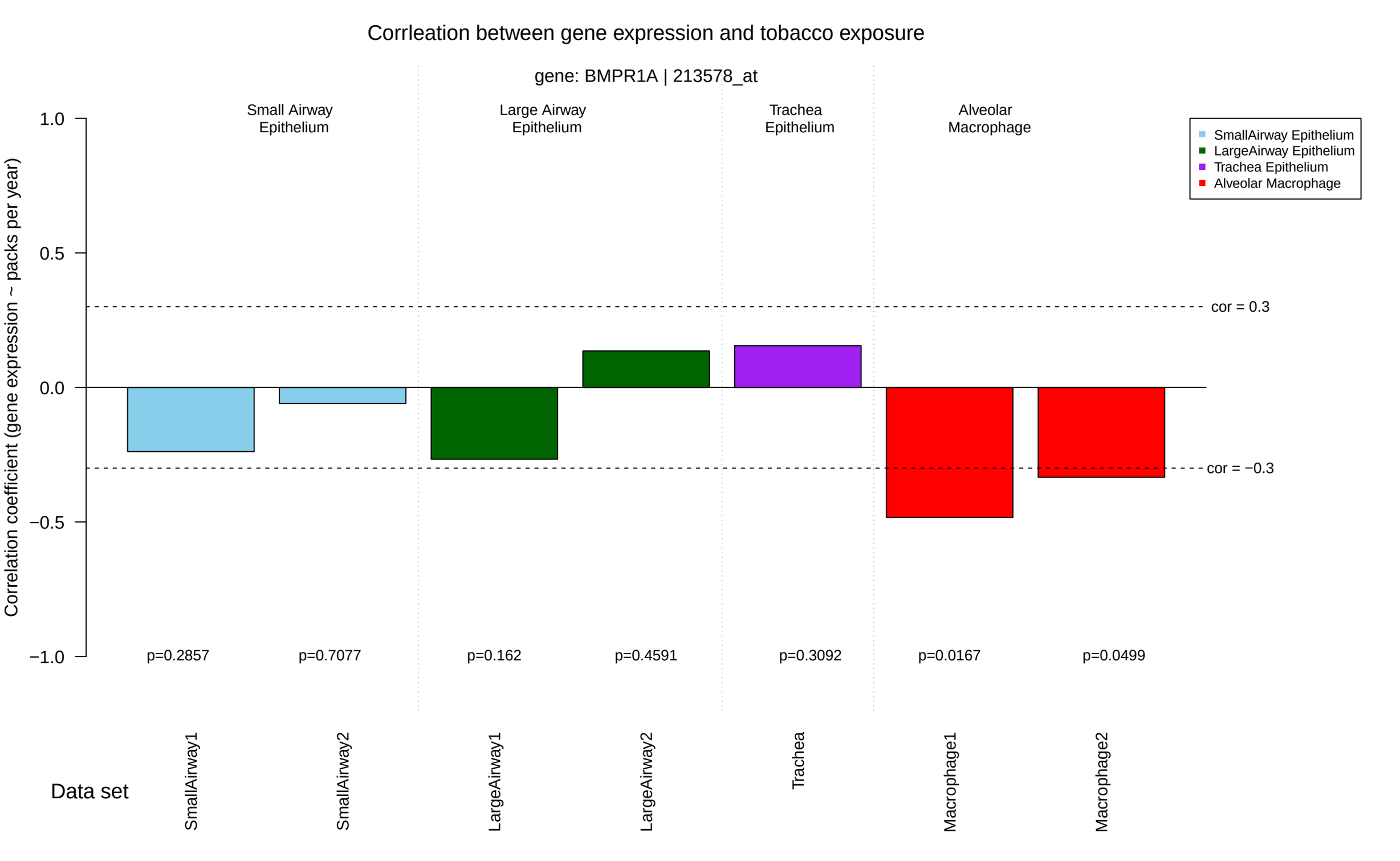

set: LargeAirway1 (p=0.0138)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8304)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.487)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0058)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0038)
BMPR1A | 213578_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMPR1A | 213578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2857
p=0.7077
p=0.162
p=0.4591
p=0.3092
p=0.0167
p=0.0499
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2